T025 historical <analysis>

Project: Real synthetic CAD
Analysis: Deck reference · frozen revision 4
Run completed: 2026-10-15T04:02:06.898731+00:00
Report generated: 2026-10-15T04:02:11.039045+00:00
Display units: mm-N-MPa

Report 419e5caa-a50b-42de-83ff-4d4f0618757b
Run efaf65d1-60fc-4627-b2e4-cbf2b7329c8c
Input SHA-256 cbd98d9934268cdba3fbffca8e8d0e982c8cf433abf9b03433241e483cb4de7d

Scope, warnings and assumptions

> Same undeformed physical coordinates and explicit element/body/material sides in both retained affine C3D10 meshes. Quadratic displacement; tensor interpolation precedes invariants, with the declared unaveraged/partitioned-smoothed mode. Changes are compared minus reference; relative change is absolute change / absolute reference, unavailable at or below the disclosed SI reference floor. Floors are display conventions, not numerical acceptance tolerances. Raw native extrema are separate moving locations. One arbitrary comparison cannot establish convergence, accuracy, stability or safety.

stress_singularity · all bodies · unassessed: Peaks near supports, reentrant corners or idealized load/bond edges may increase with refinement. No automatic singularity detection is claimed. Compare fixed physical locations away from these regions; a lower smoothed peak is not evidence of convergence or a design allowable.

yielding · 6f33a3928358ca77193419343abe0d6296faa1302201ba2451e64ad24f8716ef:b1 · not triggered: Compare raw integration-point von Mises with the frozen supplied ductile yield. Yield exceedance does not model plastic deformation; missing yield cannot establish a factor. No fatigue, brittle failure, buckling or code check is implied.

small_deformation · 6f33a3928358ca77193419343abe0d6296faa1302201ba2451e64ad24f8716ef:b1 · not triggered: Maximum native displacement / largest body extent is a 1% screening heuristic. Exceedance requires review of small-deformation linear elasticity. Passing this heuristic does not bound local strain/rotation or certify validity; display exaggeration is irrelevant.

stress_singularity · all bodies · unassessed: Peaks near supports, reentrant corners or idealized load/bond edges may increase with refinement. No automatic singularity detection is claimed. Compare fixed physical locations away from these regions; a lower smoothed peak is not evidence of convergence or a design allowable.

yielding · 6f33a3928358ca77193419343abe0d6296faa1302201ba2451e64ad24f8716ef:b1 · not triggered: Compare raw integration-point von Mises with the frozen supplied ductile yield. Yield exceedance does not model plastic deformation; missing yield cannot establish a factor. No fatigue, brittle failure, buckling or code check is implied.

small_deformation · 6f33a3928358ca77193419343abe0d6296faa1302201ba2451e64ad24f8716ef:b1 · not triggered: Maximum native displacement / largest body extent is a 1% screening heuristic. Exceedance requires review of small-deformation linear elasticity. Passing this heuristic does not bound local strain/rotation or certify validity; display exaggeration is irrelevant.

Linear static, small-deformation and supplied material assumptions apply. No automatic design-safety conclusion is generated.

Yield assessment does not assess fatigue, brittle failure, buckling or plasticity. Stress singularities require engineering review.

Passing equilibrium does not establish convergence, stability or physical validity. Target-laptop validation and remaining external references are pending; bounded M3 approval does not certify this design.

Figures are light report derivatives of retained verified data. Hidden bodies, clipping, smoothing and deformation do not alter all-body native extrema or warnings.

Selected figures and setup

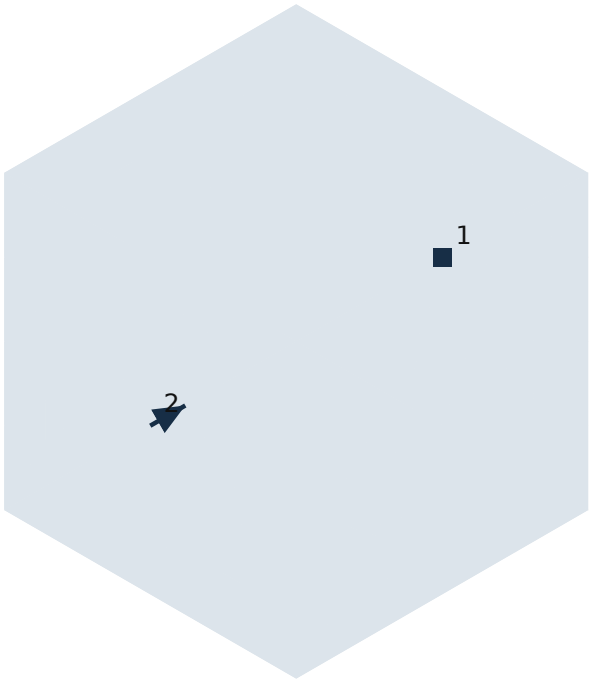
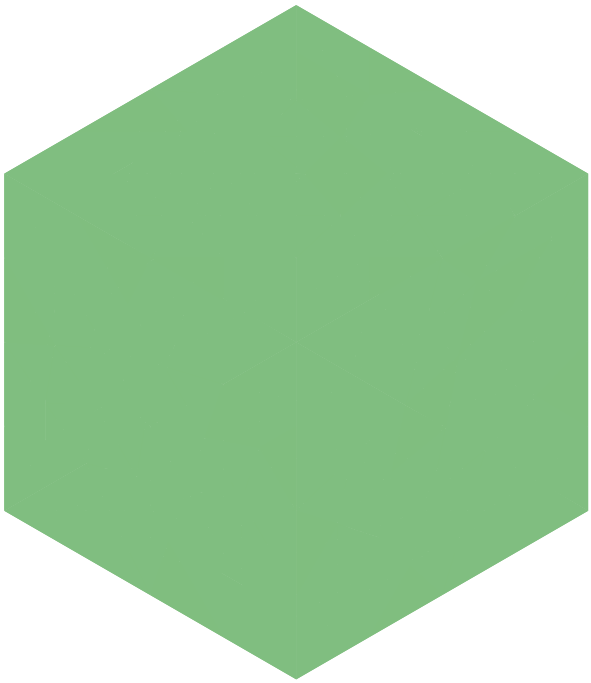


Figure 1. Frozen loads and supports on undeformed geometry. Arrows indicate global direction only; squares indicate fixed supports. Marker numbers resolve to the setup table; arrow lengths do not encode magnitude.



Stress σxx (MPa): -0.0012 to -0.0008

Colour limits clamp display only; raw extrema remain independent.

Figure 2. Stress σxx; global Cartesian; element-node boundary; Equal-weight arithmetic mean of incident element-node tensors at the same mesh node, partitioned by body and material; all incident volume elements contribute once. Invariants are calculated after tensor averaging. No spatial merging or FRD stress. Display derivative only; smoothing can hide element jumps and is not convergence evidence.. Deformation ×50; visible bodies 6f33a3928358ca77193419343abe0d6296faa1302201ba2451e64ad24f8716ef:b1. Flat triangle colour samples use the vertex-scalar mean; no numerical extrema are read from pixels. Camera and SI limits are retained in the report model.

Geometry and source units

Fact	Frozen value
Geometry revision	28cda638-cf9d-4f8d-8d6a-f2e614db9827
Source length unit	m
Bounds (mm)	[0, 0, 0, 1000, 1000, 1000]
Dimensions (mm)	[1000, 1000, 1000]
Bodies / faces	1 / 6
Geometry SHA-256	6f33a3928358ca77193419343abe0d6296faa1302201ba2451e64ad24f8716ef

Material properties

Body / material	E (MPa) / Poisson ratio	Density (tonne/mm³)	Yield (MPa)
6f33a3928358ca77193419343abe0d6296faa1302201ba2451e64ad24f8716ef:b1 / Synthetic steel	210000 / 0	7.85e-09	250

Material provenance

Body	Detail	Source / assumption
6f33a3928358ca77193419343abe0d6296faa1302201ba2451e64ad24f8716ef:b1	Grade / condition	Test only / Analytical fixture
6f33a3928358ca77193419343abe0d6296faa1302201ba2451e64ad24f8716ef:b1	Temperature / assumptions	Room-temperature fixture / Synthetic isotropic reference; no certified allowable.
6f33a3928358ca77193419343abe0d6296faa1302201ba2451e64ad24f8716ef:b1	Library / original units	None / None / mm-N-MPa
6f33a3928358ca77193419343abe0d6296faa1302201ba2451e64ad24f8716ef:b1	Independent test definition	No URL supplied; accessed not supplied

Loads and supports

Assignment	Immutable targets	Physical value / direction
fixed / fixed	6f33a3928358ca77193419343abe0d6296faa1302201ba2451e64ad24f8716ef:b1:f1	All translations fixed
pressure / pressure	6f33a3928358ca77193419343abe0d6296faa1302201ba2451e64ad24f8716ef:b1:f2	Pressure MPa: 0.001

Setup figure markers

Marker	Assignment / kind	Position (mm) / global direction
1	a2a2a9e2-3f5c-4965-ba74-44e2273cfe83 / fixed	[0, 500, 500] / [0, 0, 0]
2	7b637161-dd62-4a33-88f0-598165b6c3d4 / pressure	[1000, 500, 500] / [-1, 0, 0]

Bond review and gaps

Connection	Master / slave	Approved gap / review
None	No bonded connections	Single body or disconnected setup as frozen

Mesh and solver

Fact	Frozen value
Element / node count	C3D10 / 100 elements / 231 nodes
Quality	{"schema_version": 1, "accepted": true, "minimum_jacobian_m3": 0.031249999999996447, "minimum_mean_ratio": 0.639542072978345, "bad_elements": [], "warning_elements": [], "all_affine": true, "policy": {"jacobian_floor_m3": 1e-15, "mean_ratio_floor": 0.05, "mean_ratio_warning": 0.2, "subdivision_depth": 3, "method": "cubic tetrahedral Bernstein
	certificate with outward rounding margin"}}
Mesh controls	{"schema_version": 1, "geometry_revision_id": null, "global_size_m": null, "source_global_size": null, "source_units": null, "local_sizes": [], "curvature_elements": 16, "recovery_enabled": false}
Solver	CalculiX 2.23
Solver binary SHA-256	d55622f40e0435e738a86cbd364d525fff31524212ecca17d71d4c15ea2935f5
Original solve CPU/RAM/thread inventory	Not recorded in this legacy local run; resource policy and measured usage are retained in run artifacts.
Native build SHA-256	917a915f2aa78656131eeba825c13dece6cf81834d786a317d7dc9e6ef6db74d
Numerical method	Linear static / small deformation / global Cartesian / C3D10; no stabilization

Raw native extrema — all bodies

Field / location	Minimum	Maximum / identity
Displacement magnitude (mm); node	0; node; node 1; element None; IP None; body None; material None; position [0, 0, 0] mm	4.761905e-06; node; node 5; element None; IP None; body None; material None; position [1000, 0, 0] mm

Field / location	Minimum	Maximum / identity
Von Mises stress (MPa); integration_point	0.001; integration_point; node None; element 118; IP 4; body 6f33a3928358ca77193419343abe0d6296faa1302201ba2451e64ad24f8716ef:b1; material M1; position [909.3085, 38.86779, 456.8136] mm	0.001; integration_point; node None; element 132; IP 1; body 6f33a3928358ca77193419343abe0d6296faa1302201ba2451e64ad24f8716ef:b1; material M1; position [948.1763, 241.2459, 577.371] mm
Displacement ux (mm); node	-4.761905e-06; node; node 5; element None; IP None; body None; material None; position [1000, 0, 0] mm	0; node; node 1; element None; IP None; body None; material None; position [0, 0, 0] mm
Displacement uy (mm); node	-4.549493e-21; node; node 22; element None; IP None; body None; material None; position [1000, 0, 250] mm	9.568853e-21; node; node 79; element None; IP None; body None; material None; position [1000, 875, 875] mm
Displacement uz (mm); node	-8.157021e-21; node; node 7; element None; IP None; body None; material None; position [1000, 1000, 0] mm	2.352295e-21; node; node 23; element None; IP None; body None; material None; position [1000, 0, 750] mm
Stress σxx (MPa); integration_point	-0.001; integration_point; node None; element 85; IP 1; body 6f33a3928358ca77193419343abe0d6296faa1302201ba2451e64ad24f8716ef:b1; material M1; position [280.0724, 838.3979, 733.1534] mm	-0.001; integration_point; node None; element 85; IP 1; body 6f33a3928358ca77193419343abe0d6296faa1302201ba2451e64ad24f8716ef:b1; material M1; position [280.0724, 838.3979, 733.1534] mm
Stress σyy (MPa); integration_point	-1.941165e-18; integration_point; node None; element 118; IP 4; body 6f33a3928358ca77193419343abe0d6296faa1302201ba2451e64ad24f8716ef:b1; material M1; position [909.3085, 38.86779, 456.8136] mm	3.307091e-18; integration_point; node None; element 178; IP 3; body 6f33a3928358ca77193419343abe0d6296faa1302201ba2451e64ad24f8716ef:b1; material M1; position [941.0814, 386.5374, 532.6422] mm
Stress σzz (MPa); integration_point	-1.990566e-18; integration_point; node None; element 114; IP 2; body 6f33a3928358ca77193419343abe0d6296faa1302201ba2451e64ad24f8716ef:b1; material M1; position [913.6271, 551.8237, 965.4508] mm	3.259733e-18; integration_point; node None; element 174; IP 4; body 6f33a3928358ca77193419343abe0d6296faa1302201ba2451e64ad24f8716ef:b1; material M1; position [948.1763, 965.4508, 620.922] mm
Stress σxy (MPa); integration_point	-2.922887e-18; integration_point; node None; element 114; IP 1; body 6f33a3928358ca77193419343abe0d6296faa1302201ba2451e64ad24f8716ef:b1; material M1; position [801.8237, 663.6271, 965.4508] mm	3.649861e-18; integration_point; node None; element 132; IP 1; body 6f33a3928358ca77193419343abe0d6296faa1302201ba2451e64ad24f8716ef:b1; material M1; position [948.1763, 241.2459, 577.371] mm

Field / location	Minimum	Maximum / identity
Stress σyz (MPa); integration_point	-1.152112e-18; integration_point; node None; element 165; IP 4; body 6f33a3928358ca77193419343abe0d6296faa1302201ba2451e64ad24f8716ef:b1; material M1; position [961.1322, 948.1763, 159.7898] mm	1.64575e-18; integration_point; node None; element 160; IP 1; body 6f33a3928358ca77193419343abe0d6296faa1302201ba2451e64ad24f8716ef:b1; material M1; position [948.1763, 168.697, 952.225] mm
Stress σzx (MPa); integration_point	-2.692038e-18; integration_point; node None; element 180; IP 4; body 6f33a3928358ca77193419343abe0d6296faa1302201ba2451e64ad24f8716ef:b1; material M1; position [965.4508, 142.5152, 38.86779] mm	3.351433e-18; integration_point; node None; element 175; IP 1; body 6f33a3928358ca77193419343abe0d6296faa1302201ba2451e64ad24f8716ef:b1; material M1; position [961.1322, 38.86779, 370.4407] mm
Principal stress σ1 (MPa); integration_point	-7.60099e-19; integration_point; node None; element 144; IP 2; body 6f33a3928358ca77193419343abe0d6296faa1302201ba2451e64ad24f8716ef:b1; material M1; position [461.4968, 763.0727, 948.1763] mm	3.916902e-18; integration_point; node None; element 178; IP 3; body 6f33a3928358ca77193419343abe0d6296faa1302201ba2451e64ad24f8716ef:b1; material M1; position [941.0814, 386.5374, 532.6422] mm
Principal stress σ2 (MPa); integration_point	-2.611433e-18; integration_point; node None; element 118; IP 4; body 6f33a3928358ca77193419343abe0d6296faa1302201ba2451e64ad24f8716ef:b1; material M1; position [909.3085, 38.86779, 456.8136] mm	2.179277e-18; integration_point; node None; element 132; IP 1; body 6f33a3928358ca77193419343abe0d6296faa1302201ba2451e64ad24f8716ef:b1; material M1; position [948.1763, 241.2459, 577.371] mm
Principal stress σ3 (MPa); integration_point	-0.001; integration_point; node None; element 85; IP 1; body 6f33a3928358ca77193419343abe0d6296faa1302201ba2451e64ad24f8716ef:b1; material M1; position [280.0724, 838.3979, 733.1534] mm	-0.001; integration_point; node None; element 85; IP 1; body 6f33a3928358ca77193419343abe0d6296faa1302201ba2451e64ad24f8716ef:b1; material M1; position [280.0724, 838.3979, 733.1534] mm

Yield assessment — frozen supplied strength / raw integration-point von Mises

Body / material	Yield (MPa)	Factor / critical location
6f33a3928358ca77193419343abe0d6296faa1302201ba2451e64ad24f8716ef:b1 / Synthetic steel	250	250000; integration_point; node None; element 132; IP 1; body 6f33a3928358ca77193419343abe0d6296faa1302201ba2451e64ad24f8716ef:b1; material M1; position [948.1763, 241.2459, 577.371] mm

External reactions about global origin

Body / constrained nodes / DOFs	Force (N)	Moment (N·mm)
6f33a3928358ca77193419343abe0d6296faa1302201ba2451e64ad24f8716ef:b1 / 37 / 111	[1000, -7.105239e-14, -9.978287e-15]	[1.400531e-10, 500000, -500000]

Force and moment equilibrium

Balance	Residual	Reference	Criterion / limit	Outcome
force (N)	[-1.477929e-12, -6.698779e-14, 6.129549e-15]; norm 1.479459e-12	1000	relative; relative 1.479459e-15; allowed 5	Passed
moment (N·mm)	[1.592701e-10, -0.001962891, 0.0008496103]; norm 0.002138874	1000000	relative; relative 2.138874e-09; allowed 5000	Passed

Equilibrium method and actions

Fact	Value
Method	Frozen physical load magnitudes plus nonzero prescribed-component actuator reactions; moments about global origin; external constrained span only; internal ties/MPC/free-node RF excluded
Reaction method	DAT RF minus written equivalent CLOAD/pressure/consistent gravity, constrained global components only; unique nodes/DOFs per body; internal ties excluded
Moment origin (m)	[0, 0, 0]
Applied force (N)	[-1000, 4.064594e-15, 1.610784e-14]
Applied moment (N·mm)	[1.921697e-11, -500000, 500000]
Stress / extrema method	linear_C3D10_extrapolation_from_native_DAT_IP; element-owned tensors, then invariants; default unaveraged, optional body/material-partitioned smoothing; Displacement: all native DAT nodes; von Mises/yield factor: all raw DAT integration points, including interior

Physical refinement comparison — no convergence certification

Sample / body / position (m)	Reference / compared (mm)	Signed / absolute change	Relative change
Fixed physical sample 0 / 6f33a3928358ca77193419343abe0d6296faa1302201ba2451e64ad24f8716ef: b1 / [0.25, 0.25, 0.25]	-1.190476e-06 / -1.190476e-06	2.728123e-13 / 2.728123e-13	2.291623e-07
Fixed physical sample 1 / 6f33a3928358ca77193419343abe0d6296faa1302201ba2451e64ad24f8716ef: b1 / [0, 0, 0]	0 / 0	0 / 0	Unavailable

Comparison provenance and moving raw peaks

Fact	Value
Compared run	167c1a9e-89ed-473a-afb3-b351fde4a1c4
Compared result SHA-256	51bed28158e66b780ef45802d617f0a5fd6f97641ab07288206260cc2344393e
Physics SHA-256	440dd1ca0534c5f6b3680d9a600876be0d75f0db8d18f98f46ee32da5669fef2
Method / denominator floor	Same undeformed physical coordinates and explicit element/body/material sides in both retained affine C3D10 meshes. Quadratic displacement; tensor interpolation precedes invariants, with the declared unaveraged/partitioned-smoothed mode. Changes are compared minus reference; relative change is absolute change / absolute reference, unavailable at or
	below the disclosed SI reference floor. Floors are display conventions, not numerical acceptance tolerances. Raw native extrema are separate moving locations. One arbitrary comparison cannot establish convergence, accuracy, stability or safety.; 1e-12 m
Reference / compared raw peaks	-4.761905e-06 / -4.761905e-06
Raw peak semantics	Native peaks may move between meshes; physical samples and moving extrema remain separate.

User-written conclusion

Attributed to the report author; this text is not a computed engineering finding.

User conclusion <script>alert('x')</script> & review required.

Reproducibility IDs

schema_version: 1

job_id: efaf65d1-60fc-4627-b2e4-cbf2b7329c8c

analysis_id: 5f91b435-a62c-4844-8c2c-dc276c63fb31

analysis_revision: 4

geometry_revision_id: 28cda638-cf9d-4f8d-8d6a-f2e614db9827

input_sha256: cbd98d9934268cdba3fbffca8e8d0e982c8cf433abf9b03433241e483cb4de7d

results_sha256: f14da888cedeab8bcfb4213fba657a1a2d267987353c1fac602ad4a42bba124c

mesh_sha256: dc8414afbdd9c8b9aff196867215cd75ceba3f3895ec634933e248a94f32921e

deck_sha256: c5c5efc64b124e901ad9cf167c8b559425719b81be84dccee9877a7b8a700e6b

dat_sha256: f2e8c1f11e5cf97efefb5989a3fc927e687e6f044fcdb9f5864afb669cd56cb0

services/worker/worker/postprocess/view.py: 7ea99bb6c7c9daf7277201e6779148eba316d236c0699a9629348e8900124a8c

services/worker/worker/postprocess/inspection.py: 2671965a0e9881217553f96e8389289d24f389b7b002569b370fe92a01486f08

services/worker/worker/postprocess/comparison.py: 8970cbe6b7e9686a77176bc4d8ae432217276bf4f2f57577c55c837941d8a3aa

services/api/app/domain/comparisons.py: 7031bcf9a45c54cef209210382718a5057449d41e29fded6bab882316e389bc0

services/api/app/domain/refinement.py: b63ca772fc862a035dda9f2c3f63d06326fa3cbda50f093e5efae4b4c258a996

services/worker/worker/postprocess/fields.py: 3201afd7ad773d0865f77b992e941c5264484d87490964b84866ff87e718c2cf

services/worker/worker/postprocess/tensors.py: 9dc5efb68ebfafc05d5f964019548b801fdcad8706032554334aa0c537767584

services/worker/worker/meshing.py: 1d71f8f57bd020f96804f284f4be1d2a0fd51bf89d0bc529f48cd1aad2a52e9e

services/worker/worker/actions.py: 1a8edff017a1397b45fa667d9056a2b5e9ebf250248812928855f64dec460e9b

services/worker/worker/directions.py: 8de17d5705b81b651a5077ddf3b80e3b29a13b9f453af1c79fc2e82db06e4592

services/worker/worker/calculix/results.py: da43f91763e9880a6e8728906202f1394d7b89a31af61e6a6f163c1190c29609

services/api/app/jobs/service.py: 638d211f21fa6e9d8b4dcce0e5a6fa5b5cace3c1ce1e2eee4e3131266965ae1e

services/api/app/domain/results.py: c4c0b9b18b090bea8352ea59a2a09776d3b20783598f81c0db6b6ce18706b3ad

version: report-v1

source_commit: 9cb069ac99394c18e319eed2a81e2a425259fcde

source_dirty: false

generator_python: 3.12.14

generator_platform: Linux-6.18.44-x86_64-with-glibc2.41

generator_cpu_count: 3

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chromium_binary_sha256: 4fa61f0ba0b4f89b7bec242c32ca16d3bccf69dbfef0651924a9ba8fede5f9d6

node_binary_sha256: bc17c508ffeed0ec622934f9b7fa72f8e78da65350e63c3eceb56fa688aa5e12

services/api/app/domain/reports.py: 45a4b8349f9ae22f3bb63d50f7ae8acce9e8129136a6969f2cbdfd8274789ca0

services/api/app/domain/report_service.py: 1e417c01670abd940a3d73f22e6555d96095dbfa88db782c6efc6380b42cd776

services/worker/worker/reports/model.py: adcb4b46695ab4ddad618402cd8d6bb613332fa0a2c3ddd97d6227d0caf7dd44

services/worker/worker/reports/figures.py: bedc9575b4b72309897e2eba8fc0664ac04e84b7de74615e1d595e29eca19047

services/worker/worker/reports/html.py: 03777a92c97a5012e174f71d0b1c8f6b448bedbd466d0f3cc1fc15dd5f10ff71

services/worker/worker/reports/render.py: 02f5ed8df22732bc9a458eee8f644315d923c347ba7a03537b1c46becd5f1ecc

scripts/reports/pdf.mjs: 125def705cc32a64a6ace989b65a573dac6af6de54fd02c1ccd11064ea65c123

package-lock.json: feb1470637caf3d65d9af3d08e89ca777ca097f49b055845452b610917c45fba

uv.lock: ba9322c310e61bee9b6b005a8140ccc313123e00d76ab5a6b1a023df0acdd0fc

Download the immutable report-model JSON for complete original inputs, view definitions, camera, quality and tool policy. Its SHA-256 and this PDF’s SHA-256 are in the report record.